AI Powered Stock Market Insight Generator for the Colombo Stock Exchange

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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ABSTRACT

Over the past few years, the Colombo Stock Exchange (CSE) has witnessed a significant growth in investor participation driven by rising number of young and first-time investors entering the market. This growth reflects rising interest in financial investments, many beginner investors face difficulty to understand market conditions and assess potential risks due to the complicate structure of financial data. The lack of analytical tools that are accessible interprets financial data such as market news, external economic events and corporate disclosures which makes decision making difficult for beginner investors. Hence, investment decisions are often made based on inaccurate or misinterpreted information which increases the risk of losses.

Stock market analysis platforms in the present mainly rely on historical price variations and traditional technical indicators with minimal ability to incorporate qualitative components such as market sentiment data and effects caused by events. In addition, many of these tools are created for experienced investors and financial analysts, which makes them difficult for beginners to understand and use effectively. This drawback demonstrates a critical research gap in developing accessible systems that integrate unstructured information, real time financial events and interpretable insights to guide users towards well informed decisions for wider range of investors.

The proposed work aims to integrate sentiment analysis and event driven financial insight modeling to evaluate financial news, market announcements and investor sentiments related to Colombo Stock Exchange. By implementing machine learning, natural language processing and financial data analytics the system intends to generate transparent market insights that helps investors to understand better about the market trends and potential risks. The proposed system is likely to enable to support more informed decision making for investors while enhancing financial literacy and facilitating the development of smart financial analysis tools for developing markets

Keywords: Financial Sentiment Analysis , Event Driven Financial Modeling, Artificial Intelligence In Finance, Stock Market Analysis , Colombo Stock Exchange.

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1.INTRODUCTION
1.1 Background & context

Financial markets act a key role in modern economies by enhancing capital formations and enabling investment opportunities for public. With the rapid growth of technologies and online trading platforms, engagement in stock markets has grown substantially over the past ten years. Specifically, the Colombo Stok exchange has witnessed a notable amount of growth in investor participation, with rising number of young and first-time investors entering into the market. While this expansion highlights a positive development in financial inclusion and economic participation, it has introduced new barriers related to investment-based decision making and financial knowledge.
Investing in stock market requires the capability to interpret complicated financial information including the historical price trends, financial reports, macroeconomic indicators , announcements made from the corporate sector and global economic events. In contrast, many beginner level investors lack the analytical knowledge essential to analyze these sources effectively. Therefore, investment decisions are often affected by misleading information, speculation or market analysis relation driven emotionally rather than systematic analysis.

Studies has shown that investor sentiment and public opinion can an considerably impact financial markets. For example, research has shown that collective mood and social media sentiment information can be used as measures of stock market movements and investor's behavior. [1] Likewise, media coverage and financial news have influenced investor's perspectives and trading decisions by influencing market sentiment and projections.[2]. These observations highlight the importance of assessing qualitative data such as news, financial reports, articles when studying stock market's behavior.

Recent developments in artificial intelligence and machine learning has helped the development of models to be capable for evaluating massive datasets and extracting useful insights. Deep learning techniques are frequently utilized financial prediction problems including both financial time series analysis and data from news sources [2] [3]. Particularly, studies have showed that deep learning models can extract significant indicators from financial news and enhance stock market prediction accuracy by combining textual data along with numerical data [4].

Along with sentiment analysis, event driven approaches have likewise drawn attention in financial studies. Important financial based events and their possible impacts on market behavior will be identified and evaluated by Event driven stock prediction models. These techniques use natural language processing techniques derive relevant financial events from text sources and include them into predictive models[5]. These techniques show the growing importance of incorporating numeri financial data with unstructured text information in order to understand market dynamics in a better way.

Although these advancements, various established stock market analysis platforms mainly focus on numeri factors such as trading volumes, historical prices and technical graphs. These platforms require a considerable level of financial knowledge and may not give clear explanation of root causes modulating the market behavior. Therefore, first time investors may face difficulties to predict complex market indicators and make investment decisions.

Prompted by these challenges, this study proposes the development of an AI Powered Stock Market Insight Generation System that incorporates various data sources and analytical methods to assist investment-based decision making. The proposed system integrates finial news processing, sentiment analysis, event detection techniques and financial market data analysis to generate understandable insights related to companies that are listed to Colombo Stock Exchange. By combining, both structured financial data and unstructured text data, the system targets to find meaningful patterns, market signals and relevant factors that affects the stock market price movements. Also, the system will present this information in an interactive and user-friendly interface, which enables the investors to understand better about the potential risks ahead, market trends and influencing factors which affects the stock performance. With this approach, the proposed study intends to redefine complicated financial data into well defined, interpretable insights that enhance financial understanding and support more investment decisions

1.2 Literature Review

Over the past years, the implementation of Artificial Intelligence and data driven methods in financial markets has raised notable level of attention. Researchers have found multiple approaches to predict stock market movements and generate financial insights using numeric data, qualitative data and investor sentiment.

Prior studies on relationship between financial markets and sentiment data has highlighted that public's mood and investor's perception influence stock market behavior. As Bollen, J., Mao, H., & Zeng, X have examined large amount of social media data and found that public mood collectively could be used as an indicator to predict trends in stock market[1].

With the rapid growth of digital financial information sources, text-based analysis techniques have become important in stock market prediction research. Schumaker and Chen have analyzed the use of articles based on financial news for stock prediction by implementing text-based analysis techniques to gain relevant information from news data[2]. Their research shows that financial news contains beneficial data which can improve market prediction models when integrated with traditional financial indicators.

Later studies have focused on incorporating machine learning and deep learning methods to predict financial data. Fischer and Krauss implemented long short term memory networks to financial time series data and has shown that deep learning models achieve superior results compared to traditional approaches in some market predictions tasks.[3] Likewise, Vargas, M. R., de Lima, B. S. L. P., & Evsukoff, A. G. have presented a deep learning framework which uses financial news as input for stock market prediction demonstrating that text based data can improve the accuracy of prediction when used with historical market data.[4]

Also, event driven approaches have also become an important area of research in financial analytics field. Ding, X., Zhang, Y., Liu, T., & Duan, J have presented a deep learning-based event driven model which identifies financial events from text-based data and integrates them with stock prediction tools.[5]Their study emphasizes the importance of understanding market events and their influence on stock price movements. Additionally, to individual predictive models, various research has carried out systematic reviews of artificial intelligence applications in financial markets.

Nassirtoussi, A. K., Aghabozorgi, S., Wah, T. Y., & Ngo, D. C. L. conducted an in-depth review of text mining methods used for market prediction and highlighted the importance of natural language processing in extracting useful information from financial documents and news.[6] Similarly, Cheng, W. K., Bea, K. T., Leow, S. M. H., Chan, J. Y. L., Hong, Z. W., & Chen, Y. L. have examined sentiment based, sematic based and event extraction approaches for forecasting stock, identifying key methods used to analyze text based financial data.[7]

In recent studies, Lin and Marques performed a systematic review on AI based stock prediction, demonstrating the increase in growth level of artificial intelligence methods in financial analytics[8]. Their research shows that many predictive models focus on forecasting the price, relatively small amount of systems aim to generate interpretable insights that help investors to have a better understanding on reasons behind the market movements.

Although there are improvements, various financial system which exists mainly focuses on prediction accuracy rather than presenting the data in a way that helps investors to understand the reasons behind the behavior. Majority of platforms highly depend on numerical data such as historical prices and technical metrics, whereas the influence of financial news ,sentimental data and even driven signals remains limited. In addition , current systems need financial knowledge to predict complicated analytical data making them less approachable to the first-time investors.

1.3 Research Gap

In spite the significant progress in financial market analysis and forecasting methods, various drawbacks remain in presently available research practical financial analysis system. Several researches have mainly focused on forecasting stock price movements using historical data and numeric indicators. For example: for financial time series forecasting, deep learning models such as long short-term memory networks are widely used to highlight the performance in stock market trend forecasting. [3] Nevertheless, these methods mostly reply on quantitative data and may fail to represent text-based data which influences market behavior.

Studies have also demonstrated the use of text based analytical data such as news and social media to improve stock prediction. Researchers have found that sentiment data extracted from financial news, articles and opinion from public can offer beneficial insights into investor’s behavior and market’s dynamics.[1]. Also, event driven methods are suggested to identify financial events from text-based data and integrate them with predictive models [5]. These methods show the importance of incorporating qualitative data with numerical data, many present researches mainly focus on enhancing the accuracy of the prediction rather than providing interpretable insights which explains the market movements.

Furthermore, many reviews of the usage of artificial intelligence in financial markets have emphasized that the current research concentrate on complex predictive models than designing systems that provide insights which investors can understand. Currently many existing platforms depend on technical indicators and historical price variations which can be hard for beginner investors to interpret. Beginner investors may find it difficult to understand the factors influencing the market behavior.

Therefore, a clear research gap is found in the development of integrated system than incorporates financial data analysis, sentiment data analysis and event driven insights while presenting the results in a user-friendly manner. Resolving this gap can help change complicated financial data into meaningful data which assists investors to understand market trends and make more wise decisions.

Table 1 Research Gap

Feature	Research 1	Research 2	Research 3	Existing Trading Systems	Proposed model
Text based Sentiment Analysis	✔	✔	✔	✖	✔
Event Driven Analysis	✖	✖	✔	✖	✔
Integration of Finanial news and artiles	✔	✔	✔	Limited	✔
Finanial report analysis	✖	✖	✖	Limited	✔
Intereative Visual insights for investors	✖	✖	✖	Limited	✔
Ombinin various data soures	✖	Limited	Limited	✖	✔

2.OBJECTIVE

2.1 Main Objective

To develop and design an AI Powered Stock Market Insight Generation System that incorporates financial market data, news sentiment and company reports to provide interpretable and understandable insights for investors in Colombo Stock Exchange

2.2 Specific Objective

- To combine Explainable Artificial Intelligence techniques to provide clear and transparent explanation for AI generated output and models.
- To deploy Large Language Models to interpret financial documents and generate summaries for investors.
- To develop machine learning models which are capable of identifying risk indicators, market trends from historical data.
- To collect financial market data such as trading volumes, historical share prices and financial indicators of companies under Colombo stock exchange.

3.METHODOLOGY

The below figure 1 shows how the system diagram and the implementation that's going to be taken to build the AI Powered Stock Market Insight Generation System.

Figure 1 System Diagram

3.1 Software Solution

The research concentrates on the development of an AI Powered Stock Market Insight Generation System designed to help investors to understand better about the stock behavior within the Colombo Stock Exchange. The model incorporates various analytical components which includes financial analysis, financial document forecasting ,sentimental analysis and AI influenced mechanisms to provide interpretable insights .

The system will handle with both structured financial data such as trading volumes and historical stock prices and unstructured text-based data such as financial news, financial articles and company reports. By integrating natural language processing methods with machine learning, the system determines market trends, identify signals via sentimental news, and detect factors that influence stock price movements. The results generated by the components will be visualized in an interactive manner to improve the user experience and accessibility and better understanding for investors

3.1.1 Development Process

The development of the system will follow an agile development approach. Agile technique motivates iterative development, flexibility and teamwork which provide various benefits. Working on teams can effectively adapt to change in needs and requirements since there will be regular feedback sessions, continuous development and progressive development. Agile helps to minimize potential risks and maximize the usage of resources with a competitive advantage in rapidly moving sectors by highlighting flexibility, transparency, and giving importance for improvements in every iteration. This method increases efficiency and creativity which makes it useful in fields such as financial markets.

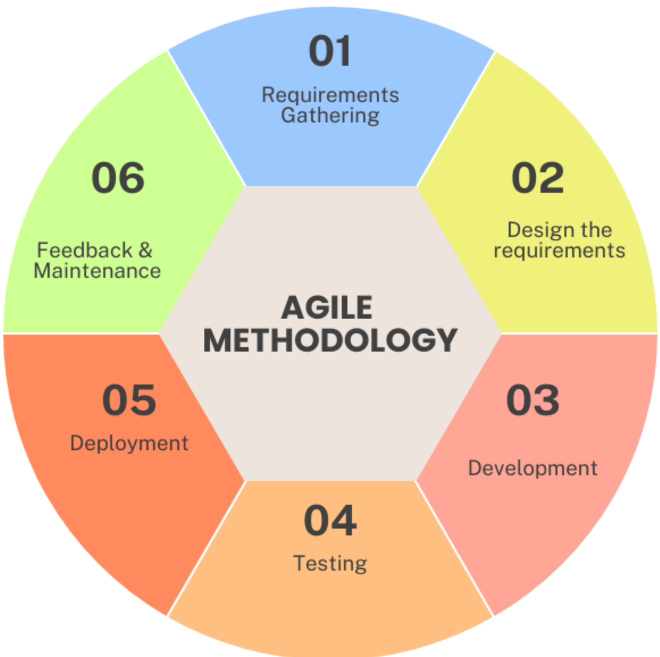

Figure 2 Agile Management

3.2 Feasibility Study

3.2.1 Technical Feasibility

The system is technically feasible due to the availability of advanced data analysis tools and AI frameworks which are capable of analyzing huge volume of text based and numerical data. Natural language processing models and machine learning libraries gives support for developing predictive models and performing sentiment analysis. These technologies facilitate effective implementation of proposed components and allow the system to process the information in real time .

3.2.1.1. Main Technologies

The below listed technological frameworks and tools are important to develop the proposed AI powered stock market insight generation system

- Natural Language processing: these techniques will be used to evaluate financial news, economic news and other text-based information in relevant to stock market. This facilitates the system to retrieve signals from sentimental news and find events that might influence the market trends
- Explainable Artificial Intelligence (XAI) : XAI will be combined to generate predictable explanations for the insights produced by the system, allowing the investors to have a better understanding about the reason behind AI generated outputs.
- Financial Data Processing : Data processing methods will be used to demonstrate and manage huge volume of financial data
- Machine learning techniques : machine learning models will be used to demonstrate historical market data to identify patterns of market trends, risk signals etc.…

Availability of the tool: technologies required for the system is available through research frameworks, open source libraries and publicly available development platforms. These tools support effectively to implement machine learning models, visually appealing components and natural language processing methods needed top develop the system.

3.2.1.2. Data Availability and Collection

Availability of data plays a significant role when developing machine learning models and analytical systems. The proposed system will use the financial news , company reports and financial datasets related to Colombo stock exchange which are available publicly.

3.2.1.3 Data Collection

As the data will be gained from external sources, it is necessary to precisely identify and select consistent and reliable financial datasets. Historical stock price data can be collected through financial information platforms, data repositories and publicly available datasets. Also, macroeconomic news and financial news announcements can be collected through news sources to support event driven analysis and sentiment news analysis.

3.2.1.4. Data Preprocessing

To ensure that the collected data is matched for machine learning data preprocessing is essential. Numerical datasets will go through the preprocessing stage, for example: removing missing values and inconsistencies etc.… for text-based data preprocessing methods like tokenization will be used. The preprocessing stage will increase the standard of the data and improve the performance of sentiment analysis.

3.2.2 Schedule Feasibility

The system will be implemented according to a project timeline which is structured to make sure completing each stage of research timely. The process will include stages like data collection, literature review, development of the model, integration etc.… Each step will be within the academic project timeline.

3.3 Requirement Gathering and Analysis

This prioritizes on understanding better about the challenges faced the investors especially first-time investors. When interpreting the stick market behavior. It identifies functionalities required to generate useful financial insights.

• **Survey with investors:** A survey will be conducted among the investors to identify the common challenges they face when evaluating stock market behavior, forecasting financial news or identifying potential risks. The data collected from the survey will help while developing the key features required and make sure that enhanced solution works for the real world's need.

• **Secondary research:** along with the survey, the research will be continued by reviewing existing studies and stock market platforms. This will ensure the drawbacks of the existing tools and gaps in the currently used platforms are demonstrated.
But incorporating, data from investor and present studies, the requirement gathering will be strong for developing the proposed system that effectively supports understanding the complicated financial market's behavior

3.4 Design and implementation

The system will be developed using a modular architecture with various integrated components like data processing , data collection , natural language processing, insight generation and machine learning module

The machine learning module will evaluate historical data to identify risks and trends in stock market behavior. Likewise, the natural language processing component will process the text-based data and news to detect sentimental news signals and highlighting events occurring in the market.

The outcomes from these components will be combined within an insight generation module which will produce forecast s of stock market behaviors. The final data will be presented in a visually appealing dashboard which allows the users to view the market trends, Ai based insights, sentiment indicators in a format with better understanding.

3.5 Testing

Testing is a crucial step to make sure that the system which is developed is correct, reliable and functional. The procedure will consist of:

- **Unit testing**: Separate components including the machine learning modules , natural language processing module will be tested independently.

- **Integration testing:** It makes sue that everything works together correctly and the data generated by the separated components are properly incorporated into the dashboard without any clashes.

- **User acceptance testing:** examines the whole system on user's point of view and makes sure that the generated insights are useful and understandable for investors.

3.6 Evaluation Plan

The system evaluation will be conducted by demonstrating the performance of machine learning and sentiment analysis components using accuracy, recall and F1score. Also, the effectives of the data generated will be evaluated by examining how well the users understand the stock market trends and behavior when using the system.

4. USER REQUIREMENTS

4.1 Stakeholder Identification

The proposed system involves various stakeholders who will interact with the system or benefit by using it.

- **Investors**

Investors are the main users of the system. They will use the system to evaluate the stock market's behavior, trends, financial news's sentiment and gain simplified outcomes that will support their investment planning and decisions.

- **Developers and Researchers**

Developers and researchers are in charge for developing, designing, implementing and maintaining the tool. They make sure that the analysis components and visualization features function properly and efficiently.

- **Financial analysts**

Experts may use the system to check the generated results and analyses the efficiency of the analytical models in forecasting the market behavior and financial information.

4.2 Functional Requirement

1. The system is designed to generate interpretable insights based on event driven analysis, financial data and sentiment signals.
2. The system will visualize sentiment indicators, market patterns and AI generated insights through a visually attractive dashboard.
3. The system aims to gather and evaluate historical stock market trends related to companies under Colombo stock exchange.
4. The system is designed to examine the news and events occurring in the financial markets to identify sentiment signals.
5. The system intends to identify market trends risk factors using machine learning techniques.
6. The system expects to generate simplified summaries of the financial documents to improve the investor's understanding.

The system allows users to explore company specific data and relevant data

4.3 Non-Functional Requirement

1. **Security**: The system makes sure the financial data is handled securely and protect the platform from third part access.
2. **Reliability**: the system intends to maintain consistency and provide accurate analytical output.
3. **Usability**: the platform aims to present a user-friendly interface which allows investors to interpret easily complicated financial information
4. **Performance**: the system will be designed to efficiently process large amount of financial data and responds within a reasonable timeframe.

5.GANTT CHART

The below chart shows the Gantt chart of the component that's going to be designed as project is planned

Figure 3 Gantt chart

6.WORK BREAKDOWN CHART

The below shown figure shows the work breakdown chart which showcases the planning and how the subtasks are broken down into smaller steps and are going to be implemented to achieve the main objective.

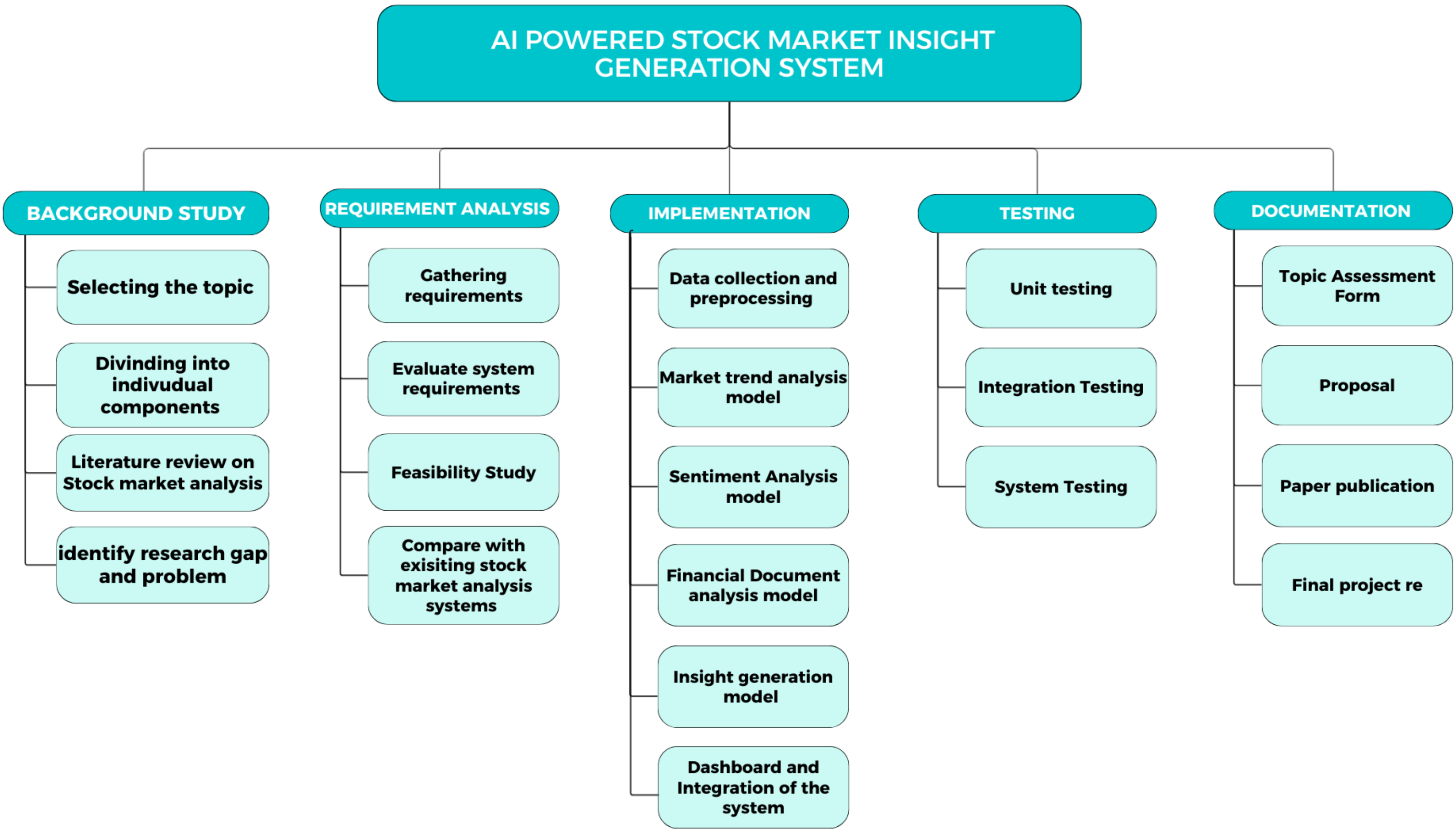

Figure 4 Work Breakdown Chart

7.BUDGET

As the proposed output is a software-based solution, there are no hardware components needed for the implementation. The main expenses are others like internet usage needed for development, testing a deploying the system etc... The estimated budget is mentioned below.

Expense type	**Cost (LKR)**	**Justification**
Internet use and Hosting	12,000	Needed for hosting web-based platform and consistent network connection while developing, testing and deploying the system.
Course related to ML	8000	To gain knowledge from online learning courses related to machine learning techniques to implement while developing the model
Publishing expense	16,000	Covers research publication or conference submission expense
Travelling charges	5,000	Needed for travelling for supervisor meetings, attending presentations and conducting physical data collection
Data collection	2,000	Cost for printing questionnaires if physical data collection takes place amongst the investors.
Miscellaneous expense	3,000	Covers minor expenses like printing documentations, reports and small project related expenses
TOTAL	**46000**	

Table 2 Budget

The expenses may differ accordingly relevant to the need

8.COMMERCIALIZATION

The AI Powered Stock Market Insight Generation system has significant commercialization potential as support for decision platform for investors. By combining sentimental new analysis, financial news analysis and event detection methods, the system changes complicated financial data into simplified, clear and interpretable insights which helps investors to understand better about the market’s behavior, trends and risks.

The model is developed to be accessible and scalable via a web-based interface, allowing incorporating with currently existing financial platforms and advisory services. By generating transparent AI based insights, the system can help investors in making more wise investment decisions while improving trust in AI tools

Additionally, the proposed solution can offer integrated service with financial institutions, fintech companies and advisory services offering an subscription-based analytics platform. Advanced predictive analytics, personalized insights may be improved in future which can increase the commercial value and the broader adaption of the system

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